Chapter 5 – Congruent Triangles Lab

You need: a ruler, a protractor, a compass, scissors and a few strips of paper or straws. You are about to find out WHY five sets of parts force two triangles to be congruent — and why a sixth set does not. Everything you build here is a theorem you will use for the rest of the year.

Lab 1 – Three Sides (SSS)

Build a triangle from three lengths. Then try to build a DIFFERENT one from the same three.

Step 1. Cut strips of 6 cm, 8 cm and 10 cm.

Step 2. Lay them into a triangle and tape the corners. Trace it.

Step 3. Take it apart and try to make a triangle of a different shape from the same three strips. Trace any you manage.

- How many genuinely different triangles could you build?

  _____

- Flipping or turning your tracing does not count as different. Why not?

  _____

- Complete: three pairs of congruent _____ force the triangles to be congruent. This is the _____ Congruence Theorem.

  _____

Lab 2 – Two Sides and the Angle Between (SAS)

Now fix two lengths and the angle BETWEEN them.

Step 1. Draw a 7 cm segment. At its left end measure a 50° angle and draw a 5 cm segment along it.

Step 2. Join the two far ends. Trace the triangle.

Step 3. Repeat on fresh paper, same numbers. Cut one out and lay it on the other.

- Did the two tracings match exactly?

  _____

- Try it again with the 50° angle NOT between the two sides. Did you get the same triangle?

  _____

- Complete: two sides and the _____ angle force congruence. Name the theorem: _____.

  _____

Lab 3 – Two Angles and a Side (ASA and AAS)

Fix two angles and one side, and see where the third vertex can go.

Step 1. Draw a 9 cm segment. Measure 40° at one end and 65° at the other, and draw both rays.

Step 2. Mark where the rays meet. Trace the triangle.

Step 3. Repeat, but this time put the 9 cm side so it is NOT between the two angles — opposite the 65°.

- Were both triangles forced, or could you build two different ones?

  _____

- What is the third angle in each case? How did you know without measuring?

  _____

- Complete: the side BETWEEN the angles is _____; a side not between them is _____.

  _____

Lab 4 – The One That Fails (SSA)

Two sides and an angle NOT between them. This is the one to be suspicious of.

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Step 1. Draw a ray from a point A, and measure a 30° angle at A. Draw the other ray long.

Step 2. On one ray mark B so that AB = 9 cm.

Step 3. Open a compass to 5 cm, put the point on B, and swing an arc across the other ray.

- How many times did the arc cross the ray? Mark every crossing.

_____

- Draw a triangle to EACH crossing. Are they congruent to each other?

_____

- Why is "SSA" not on the list of congruence theorems? Answer in one sentence.

_____

- Now redo the swing with the compass at 12 cm. How many triangles this time? What changed?

_____

Lab 5 – Right Triangles (HL)

SSA fails in general. Here is the one case where it does not.

Step 1. Draw a right angle. Make one leg 6 cm.

Step 2. Open a compass to 10 cm, put the point at the top of the leg, and swing it onto the other ray.

Step 3. Trace the triangle. Try to build a different triangle with the same leg and the same 10 cm hypotenuse.

- How many triangles are possible?

_____

- This is SSA — two sides and an angle not between them. Why does it work HERE and not in Lab 4?

_____

- Complete: in a _____ triangle, a leg and the _____ force congruence. That is the _____ Congruence Theorem.

_____

Lab 6 – The Rule Table

Fill this in from what you built. The last column is the point of the whole lab.

Parts given	Forced? (yes / no)	Theorem name, or why it fails
Three sides		
Two sides and the angle between		
Two angles and the side between		
Two angles and a side not between		
Two sides and an angle not between		
Leg and hypotenuse of a right triangle		
Three angles		

- Two rows in this table describe the SAME parts and get different answers. Which two, and why?

_____

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- ► Three angles: you have not built this one. Try it — draw a triangle with angles 60°, 60°, 60° twice, at two different sizes. What does that tell you?
